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Machine learning-based screening model for Tic disorders comorbid with attention-deficit/hyperactivity disorder in children and SHAP value interpretation: a retrospective observational study

Zilin Chen[1], Xu Wang[2], YuXin Chai[1], YuChen Hu[1], Ru Jia[1] & Fei Han[1]✉

Tic disorders (TD) are common neurodevelopmental conditions in children and adolescents, characterized primarily by motor and vocal tics, with heterogeneous symptom presentations and significant fluctuations. Concurrently, attention-deficit/hyperactivity disorder (ADHD) is a highly prevalent comorbidity of TD, often exacerbating academic and emotional difficulties. However, due to the potential masking of ADHD symptoms by TD manifestations and insufficient awareness of comorbidities, ADHD diagnosis is frequently delayed. This study aimed to develop a machine learning-based classification model to identify children with TD who have a higher model-predicted probability of ADHD comorbidity, thereby potentially aiding diagnostic accuracy and treatment planning. The research was conducted by the Pediatrics Department of Guang'anmen Hospital, China Academy of Chinese Medical Sciences, from October 2023 to October 2024. A retrospective cohort of 1,364 children with TD was included, comprising 889 with TD alone and 475 with TD comorbid with ADHD. The dataset was divided into a training set and an independent test set using a stratified 7:3 split, with the random seed fixed at 42 to ensure the balance of baseline characteristics between the two groups. 10-fold cross-validation was used for hyperparameter tuning and preliminary performance evaluation within the training set, and the final model performance was verified on the independent test set. Twenty-eight clinical features were collected, encompassing general information, family factors, academic performance, medical history, emotional status, and sleep quality. Core variables were selected using Lasso regression, and five distinct machine learning models were constructed. Feature importance was interpreted using SHAP analysis. Ultimately, seven key features were identified for algorithm development: learning initiative, guardian type, academic performance, allergic rhinitis, irritability, adenoid hypertrophy, and sleep quality. The random forest algorithm emerged as the optimal model, achieving an AUC of 0.780 (95%CI: 0.742–0.818), accuracy of 0.793, recall of 0.793, and F1-score of 0.753 on the test set; the Brier score was 0.18, indicating good calibration of the model. SHAP analysis elucidated the contributions of specific features to the model's prediction of TD with ADHD, with learning initiative (mean |SHAP| = 0.07) and guardian type (0.05) identified as the most influential predictors, followed by academic performance (0.04), allergic rhinitis (0.03), irritability (0.03), adenoid hypertrophy (0.02), and sleep quality (0.02). These findings may provide preliminary references for future screening research and support further investigation into early identification of comorbidity, pending independent validation. However, external validation is required before any clinical implementation. This is a single-center retrospective study without external validation, and the generalization ability of the model needs to be further verified.

Keywords Tic disorders, Attention-Deficit/Hyperactivity disorder, Comorbidity, Machine learning, SHapley additive exPlanations

Abbreviations

ADHD	Attention-Deficit/Hyperactivity Disorder
TD	Tic disorders
SHAP	SHapley Additive exPlanations
YGTSS	Yale Global Tic Severity Scale
SNAP-IV	Swanson, Nolan, and Pelham–IV Rating Scale Swanson
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
DCA	Decision Curve Analysis
PRC	Precision-Recall Curve
Light GBM	Light Gradient Boosting Machine
XGBoost	Extreme Gradient Boosting
MLP	Multilayer Perceptron
Lasso	Least Absolute Shrinkage and Selection Operator
OCD	Obsessive-Compulsive Disorder
ASD	Autism Spectrum Disorder

[1]Department of Pediatrics, Guang'anmen Hospital, China Academy of Chinese Medical Sciences, Beijing, China. [2]State Key Laboratory of Traditional Chinese Medicine Syndrome, The Second Affiliated Hospital of Guangzhou University of Chinese Medicine, Guangzhou, Guangdong, China. ✉email: hanfei@gamyy.cn

Tic disorders (TD) are a type of neurodevelopmental condition with onset in childhood and adolescence, characterized corely by motor or vocal tics, and exhibiting significant heterogeneity and symptom fluctuation in clinical presentation[1]. Epidemiological data indicate that the prevalence of TD in children is approximately 0.3% to 1%, with a significantly higher incidence in males than in females (male-to-female ratio about 3:1). Symptoms typically first appear between 4 and 6 years of age, peak around adolescence, and may persist into adulthood in some patients[2]. The pathophysiological mechanisms of TD have not been fully elucidated. Studies suggest that TD may be correlated with aberrant neural transmission in the cortico-striato-thalamo-cortical circuit[3], and dysregulation of the dopaminergic system is considered one of the core mechanisms[4]. Attention-deficit/hyperactivity disorder (ADHD) is another common neurodevelopmental disorder of childhood, characterized primarily by persistent inattention, hyperactivity, and impulsivity[5]. Globally, the prevalence of ADHD in children and adolescents is approximately 8.0%, with the rate in boys (10%) being twice that in girls (5%)[6]. Individuals with ADHD often experience academic difficulties, social impairments, and emotional dysregulation[7,8]. These challenges not only affect their daily lives but are also associated with an increased risk of various psychiatric disorders during the transition from childhood to adulthood[9].

Numerous clinical studies have demonstrated a significant comorbidity between TD and ADHD[10]. Epidemiological investigations indicate that approximately 50%-60% of children with TD also have comorbid ADHD, while the prevalence of TD in ADHD patients is significantly higher than in the general population[11]. This comorbidity not only associated with more severe clinical symptoms in affected children, such as learning difficulties, emotional dysregulation, sleep disturbances, self-injurious behaviors, and elimination problems[12], but is also linked to poorer long-term outcomes, including impaired social adaptation and reduced quality of life[13]. However, in current clinical practice, the recognition and diagnosis of TD comorbid with ADHD remains delayed. On one hand, the tic symptoms in children with TD may mask the core manifestations of ADHD[14]; on the other hand, insufficient understanding of the comorbidity mechanisms among some clinicians may result in underdiagnosis or misdiagnosis[15]. The high heterogeneity of clinical phenotypes further increases the difficulty in identifying TD-ADHD comorbidity and the risk of diagnostic delay, highlighting the necessity of developing a prediction model based on multi-dimensional clinical indicators[16].

Given the multifactorial and heterogeneous nature of TD-ADHD comorbidity, traditional statistical methods may be limited in capturing complex, non-linear interactions among risk factors. Machine learning has emerged as a powerful tool for developing predictive models in psychiatry, capable of integrating high-dimensional clinical data to improve diagnostic accuracy and risk stratification for neurodevelopmental disorders[17]. For instance, ensemble learning algorithms such as Random Forest and Light GBM have been successfully applied to model complex clinical phenotypes, demonstrating superior performance in identifying patterns within multifaceted data[18]. Furthermore, interpretability methods like SHAP (SHapley Additive exPlanations) have enhanced the transparency of these "black-box" models, facilitating the identification of key predictive factors and their directional contributions, which is crucial for clinical translation[19]. For instance, in predicting Tourette Syndrome (TS), researchers have developed a gradient boosting machine model based on blood biochemical markers (e.g., β2-microglobulin and serum 25-hydroxyvitamin D3), with SHAP interpretation highlighting the key roles of these biomarkers[20]. Similarly, in screening for ADHD, comparative studies of algorithms such as logistic regression, classification and regression trees (CART), and neural networks have demonstrated that the CART model can achieve relatively high discriminatory performance[21]. However, these studies have primarily focused on single disorders and often rely on laboratory tests or specialized assessments, which may limit their feasibility in primary care settings.

Therefore, we have developed a machine learning-based classification model to predict ADHD comorbidity in children with TD, specifically aiming to distinguish between TD alone and TD comorbid with ADHD. A total of 28 clinical features were collected, and diagnoses were assisted using the Yale Global Tic Severity Scale (YGTSS), Conners, and Swanson, Nolan, and Pelham-IV Rating Scale (SNAP-IV). These indicators were selected not only because the 28 clinical features comprehensively cover multiple dimensions including general information, family guardianship patterns, academic performance, medical history, allergic predisposition, emotional status, and sleep quality, but also because they systematically capture potential associated factors of comorbidity, making them particularly suitable for early predictive classification of TD comorbid with ADHD. The core innovations of this study include: (1) developing a model with potential relevance for future screening research in primary care settings based on multi-dimensional clinical indicators that can be easily obtained in routine outpatient clinics; (2) integrating LASSO feature selection, multi-algorithm comparison, clinical utility evaluation and SHAP interpretability analysis to form a complete clinical prediction model construction process. This study aims to develop and validate an interpretable machine learning-based prediction model for identifying high-risk groups of TD children complicated with ADHD, providing data support for early clinical screening. Overall, the study included 28 clinical features of 1364 TD children, screened core indicators by Lasso regression, constructed 5 machine learning models and compared their performance, and finally used SHAP analysis to explain the prediction logic of the optimal model, providing tools and basis for early screening of TD complicated with ADHD.

Methods

Study design

As illustrated in Fig. 1, this retrospective observational study was conducted in the Pediatrics Department of Guang'anmen Hospital, China Academy of Chinese Medical Sciences, spanning from October 2023 to October 2024. The TD with ADHD group consisted of children diagnosed with both TD and ADHD, while the control group comprised children diagnosed with TD only. A descriptive summary of the demographic and clinical baseline characteristics of the study population is provided in the Supplementary Table S1.

The exclusion criteria included patients with other severe psychiatric disorders, neurological diseases, or systemic conditions affecting cognitive function, as well as cases with incomplete clinical data. To develop the clinical prediction model, we focused on identifying differences in multi-dimensional indicators, including

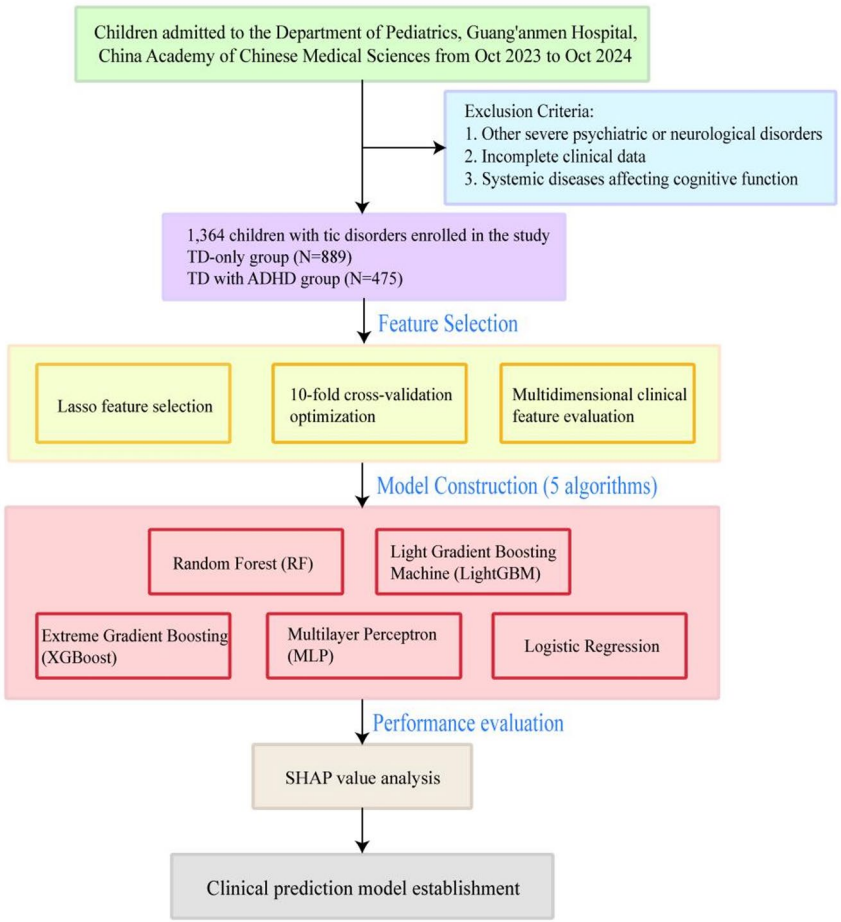


Fig. 1. Flow chart of the study design.

general information, family factors, academic performance, medical history, emotional status, and sleep quality. All data were systematically collected from patients' electronic medical records. Initially, a total of 28 features were included for analysis.

Ethical statement

This retrospective observational study was conducted in strict accordance with the ethical principles outlined in the Declaration of Helsinki. The study protocol was approved by the Ethics Committee of Guang'anmen Hospital, China Academy of Chinese Medical Sciences (Approval No.: 2023-211-KY-01). All legal guardians of the participants were fully informed of the study's purpose, procedures, and their rights, and provided written informed consent. All collected information was strictly anonymized and stored in a secure environment to ensure participant privacy and data security.

Diagnostic criteria

The Western medical diagnostic criteria for both TD and ADHD strictly adhered to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5). To enhance objectivity and consistency in diagnosis, standardized scale-based assessment tools were employed as supplementary measures for both conditions.

The YGTSS is a widely used clinical instrument for evaluating the severity of TD. It provides a comprehensive assessment across multiple dimensions, including tic frequency, intensity, complexity, interference, and overall functional impairment. The total YGTSS score is interpreted as follows: <25 indicates mild severity, 25–50 indicates moderate severity, and > 50 indicates severe severity.

The Conners' Rating Scales, completed by guardians, are behavioral assessment tools primarily used to support the diagnosis and severity evaluation of ADHD. Scores are classified as follows: 0–10 falls within the normal range, 11–19 suggests mild abnormalities, 20–29 indicates moderate abnormalities, and ≥ 30 signifies severe behavioral abnormalities requiring further clinical evaluation and intervention.

The SNAP-IV, revised based on DSM-IV criteria, is a commonly used tool for screening ADHD symptoms and assessing their severity. The total score is interpreted as follows: ≤13 is within the normal range, 13–17 indicates mild abnormalities, 18–22 suggests moderate abnormalities, and 23–27 indicates severe abnormalities.

Collection and selection of clinical features

Our study systematically collected electronic medical record data to comprehensively document multi-dimensional clinical information from children with TD. The collected data encompassed the following categories: General information (sex, age), Birth history (parity, mode of delivery, difficult labor, preterm birth, birth asphyxia, resuscitation history, low birth weight), Family factors (parental guardianship, parenting style), Academic performance (learning initiative, academic achievement), Medical history (history of common cold, febrile seizures, head trauma), Precipitating or exacerbating factors (parental discipline, poor exam performance, high stress, fright, prolonged smartphone use, extended television viewing), Constitution-related factors (frequent colds, chronic pharyngitis, chronic rhinitis, asthma history, adenoid hypertrophy), Allergic disease history (allergic rhinitis, allergic conjunctivitis, allergic purpura), TD clinical symptoms (persistent episodes, eye movements, nose movements, mouth movements, facial movements, head and neck movements, shoulder movements, upper limb movements, trunk movements, lower limb movements, throat sounds), Comorbidities (e.g., comorbid allergic rhinitis), Emotional and behavioral characteristics (irritability, timidity), Lifestyle and somatic status (dietary habits, sleep quality, bowel and urinary habits).

From a clinical definition perspective, "allergic rhinitis" is classified by etiology, "chronic rhinitis" is classified by disease course, and "comorbid allergic rhinitis" is a comorbidity diagnostic label. These three are distinct clinical concepts with independent clinical significance. From a statistical perspective, the core advantage of Lasso regression is its ability to automatically handle statistical redundancy between variables through regularization. In this study, Lasso ultimately retained only "allergic rhinitis" as a core feature, verifying that this method can effectively screen out the most predictive variables.

To ensure data quality and consistency, we implemented standardized protocols for feature collection: (1) Parental factors (parental guardianship, parenting style) were assessed via structured questionnaires completed by legal guardians during the initial clinical visit, utilizing standardized response options (e.g., "biological parents as primary guardians" vs. "other relatives/guardians"; "authoritative" vs. "permissive" parenting) to ensure objectivity. The standardized questionnaire used in this study is a routine clinical assessment content of the outpatient clinic for neurodevelopmental disorders in the Department of Pediatrics of our hospital. All included children are required to complete this set of questionnaires by their legal guardians during the initial diagnosis, which is used to assist clinical diagnosis and baseline assessment of the condition. The questionnaire data will be simultaneously entered into the electronic medical record system. This study is a retrospective analysis, which directly extracts the existing routine assessment data in the electronic medical record, and does not carry out additional data collection work for this study. (2) Variables related to prolonged smartphone use and extended television viewing were operationally defined as ≥ 2 h of daily screen time, consistent with clinical guidelines[22]. These were extracted through a dual-verification process: primarily from guardians' structured behavior reports (integrated into the routine clinical assessment form), and secondarily cross-referenced with physicians' documented observations in the electronic medical records. (3) Crucially, all aforementioned variables were collected through structured data entry templates, not via subjective interpretation during retrospective chart review. Trained research assistants provided uniform instructions to guardians for questionnaire completion, and physicians verified the clinical consistency of the responses. For the minimal cases with incomplete structured data (affecting < 3% of the total sample), the variables were excluded rather than inferred, thereby preserving data integrity and avoiding bias. All data were uniformly collected and organized by trained professionals to ensure accuracy and completeness.

As illustrated in Fig. 1, we employed the Lasso regression algorithm for feature selection. To avoid data leakage, Lasso feature selection adopts a nested cross-validation strategy: We used a stratified 70/30 train-test split. All feature selection and hyperparameter tuning were conducted only within the training set using 10-fold cross-validation; the test set was kept untouched for final evaluation. A Lasso regression model was constructed, and the regularization parameter λ was optimized through 10-fold cross-validation. The optimal λ value was determined based on maximizing the Area Under the Curve (AUC), thereby identifying predictive variables significantly associated with TD comorbid with ADHD. Notably, non-discriminatory variables (e.g., isolated history of common colds) were automatically excluded during the Lasso selection process due to their minimal contribution to model performance. For variables such as academic performance, we explicitly distinguished between "baseline academic performance prior to TD symptom exacerbation" and "academic difficulties secondary to ADHD symptoms" through retrospective medical record review. The "academic performance" and "learning initiative" indicators included in this study are all baseline data before the aggravation of TD symptoms. The time node is clearly earlier than the clinical diagnosis of ADHD through the course records in the electronic medical record, avoiding the circular argument of "using functional impairment caused by ADHD to reversely predict ADHD". This distinction avoids circular reasoning, as we focused on pre-existing academic functioning rather than ADHD-induced impairments (Table 1). To further interpret the model and evaluate feature importance, we utilized SHAP value analysis. A positive SHAP value indicates that the feature contributes to the model's prediction of a higher probability of comorbid ADHD, while a negative value suggests it contributes to a prediction of lower probability.

Model development and evaluation

Feature selection was performed using the Lasso regression algorithm to effectively mitigate overfitting. Ultimately, seven core features were identified for predictive model development. Five distinct machine learning algorithms were employed to predict the probability of being classified as TD comorbid with ADHD: Random Forest, Light Gradient Boosting Machine (Light GBM), Extreme Gradient Boosting (XGBoost), Multilayer Perceptron (MLP), Logistic Regression. The adoption of a multi-model comparison approach facilitated the evaluation of each algorithm's performance, thereby identifying the most effective technique for predicting comorbidity risk. To ensure robust model evaluation while maintaining computational efficiency, we implemented a hold-out validation approach with internal cross-validation for hyperparameter optimization. The dataset was first split into a training set (70%) and an independent test set (30%) using stratified random sampling (random seed = 42). Within the training set, we employed 10-fold cross-validation during Bayesian optimization to tune hyperparameters for each algorithm. The independent test set was held out from all model development and tuning processes and served exclusively for final performance evaluation, providing an unbiased estimate of model generalizability. AUC, accuracy and F1 value are model performance quantitative indicators, which are used to directly compare the predictive ability of different algorithms; PRC, DCA and calibration curve are clinical utility evaluation frameworks, which are used to verify the applicability of the model in actual clinical scenarios. This study uses the Youden index to determine the optimal classification threshold; to address the imbalance in sample size between the TD group and the comorbid group, a class weight adjustment strategy is adopted during model training to reduce the dominant effect of majority class samples. For logistic regression and MLP, continuous features are standardized by Z-score; tree models (random forest, LightGBM, XGBoost) do not require standardization. All binary variables are coded with 0/1; there are no multi-category variables in the core features. Bayesian Optimization is used for hyperparameter tuning, the optimization target is AUC, and the search range is determined based on literature reports and pre-experiment results. For the calculation of 95% confidence intervals (CIs) for all performance metrics, we implemented a non-parametric bootstrap resampling approach with 1000 iterations. This method involved repeatedly sampling with replacement from the test set, recalculating the performance metrics for each bootstrap sample, and then deriving the 2.5th and 97.5th percentiles as the lower and upper bounds of the 95% CI. The reliability of the models was comprehensively evaluated using four core metrics: Area Under the Receiver Operating Characteristic Curve (AUC), Precision-Recall Curve (PRC), Decision Curve Analysis (DCA), Calibration Curve. Furthermore, performance metrics including accuracy, recall, precision, and F1-score were thoroughly compared and analyzed.

Model interpretation

To gain an in-depth understanding of the model's predictive logic, we employed SHAP to interpret the best-performing Random Forest model. The importance of each feature was assessed by calculating its mean absolute SHAP value (mean(|SHAP|)). The SHAP beeswarm plot illustrates the direction and magnitude of each feature's

Core Feature	Data Source	Coding method
Learning Initiative	Guardian Standardized Questionnaire	Binary (1 = Active, 0 = Passive)
Guardian Type	Guardian Standardized Questionnaire	Binary (1 = Parental Guardianship, 0 = Others)
Academic Performance	School Report Card + Guardian Report	Binary (1 = Good, 0 = Poor)
Allergic Rhinitis	Clinical Diagnosis Record	Binary (1 = Yes, 0 = No)
Irritability	Physician Clinical Behavior Assessment	Binary (1 = Present, 0 = Absent)
Adenoid Hypertrophy	Imaging Examination Results	Binary (1 = Yes, 0 = No)
Sleep Quality	Guardian Standardized Questionnaire	Binary (1 = Good, 0 = Poor)

Table 1. Core feature details.

contribution to the model's predicted probability of TD comorbid with ADHD. Positive SHAP values indicate a contribution toward a higher predicted probability, whereas negative SHAP values indicate a contribution toward a lower predicted probability. Importantly, SHAP values reflect model attribution rather than causal effects or biological mechanisms.

Statistical analysis

Data in this study are presented as median and interquartile range [M (Q1, Q3)]. Normality of continuous variables was assessed using the Shapiro–Wilk test. Group comparisons for normally distributed data were performed using independent samples *t*-tests, while the Mann–Whitney U test was applied for non-normally distributed data. Categorical variables are expressed as frequencies and percentages, with group comparisons conducted using chi-square tests or Fisher's exact test, as appropriate. All statistical analyses were performed using R language (version 4.2.1) and relevant statistical packages. A P-value less than 0.05 was considered statistically significant. The statistical tests described in this section are only used for descriptive comparison of baseline characteristics of children in the two groups, providing clinical basis for subsequent feature screening; the core conclusion of this study is based on the prediction performance indicators of machine learning models, rather than the P-value of traditional hypothesis testing. To justify the sample size, we performed a post-hoc power analysis for the primary outcome (AUC of the Random Forest model). Assuming a moderate expected AUC of 0.75 under the null, a two-sided type I error of 0.05, and a power of 0.90, the required minimum number of positive cases (TD with ADHD) was calculated to be approximately 190 using the method of Obuchowski and McClish. Our study included 475 positive cases, which exceeds this requirement.

Results

Data processing and feature selection

A total of 1,364 children with TD were included in this study, comprising 889 cases in the TD-only group and 475 cases in the TD with ADHD group. The data processing and feature selection workflow are illustrated in Fig. 1. Ultimately, 1,364 children were included in the analysis. No significant differences were observed between the two groups in most baseline characteristics, such as sex, mode of delivery, preterm birth, birth asphyxia, or resuscitation history ($P > 0.05$). However, significant differences ($P < 0.05$) were identified in the following 14 features, suggesting their potential association with the model-predicted probability of comorbidity (Supplementary Table S1): Children in the TD with ADHD group were significantly younger and exhibited significantly higher proportions or incidences of difficult labor, inadequate parental supervision, poor learning initiative, low academic performance, history of febrile, precipitation by fright, chronic rhinitis, adenoid hypertrophy, allergic rhinitis, lower limb involuntary movements, vocal tics, irritability, and poor sleep quality compared to the TD-only group ($P < 0.05$).

Subsequently, we defined the binary outcome variable as the presence or absence of ADHD comorbidity and included 28 clinical features as initial predictive variables. To identify the most predictive variables and mitigate overfitting, Lasso regression was employed for feature selection. The regularization parameter λ was optimized via 10-fold cross-validation, with the optimal λ value (λ = 0.012) determined by maximizing the AUC (Fig. 2A). The Lasso regression path analysis showed that as λ decreased, features significantly associated with the outcome were gradually incorporated into the model (Fig. 2B).

Ultimately, Lasso regression identified seven core features significantly associated with TD comorbid with ADHD: learning initiative, guardian type, academic performance, allergic rhinitis, irritability, adenoid hypertrophy, and sleep quality. The importance of these features within the LASSO model was quantified by the absolute values of their coefficients (|β|), yielding the following ranking: Learning initiative (|β| = 0.326) > Guardian type (|β| = 0.271) > Academic performance (|β| = 0.218) > Allergic rhinitis (|β| = 0.185) > Irritability (|β| = 0.179) > Adenoid hypertrophy (|β| = 0.103) > Sleep quality (|β| = 0.098) (Table 2). At the optimal λ, the LASSO model itself achieved an AUC of 0.82 (95% CI: 0.78–0.86) in 10-fold cross-validation, confirming its robust performance for comorbidity prediction. The precision-recall curve (PRC) of the Lasso model was

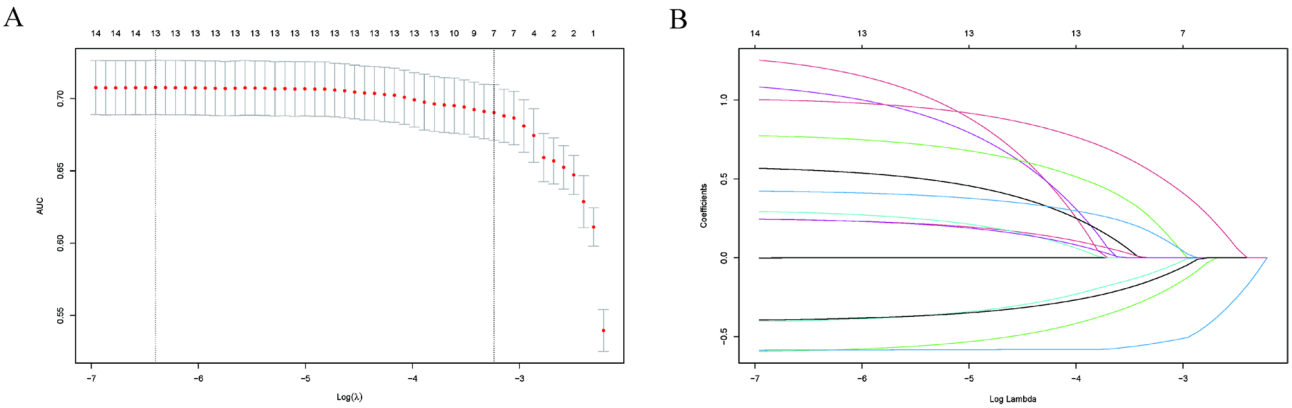


Fig. 2. (A) Selection of the optimal λ value through 10-fold cross-validation. (B) Coefficient path plot of the Lasso regression model.

Feature	Coefficient	Standard error	*P*-value
Learning initiative	-0.326	0.089	< 0.001
Guardian type	-0.271	0.094	0.004
Academic performance	-0.218	0.076	0.005
Allergic rhinitis	0.185	0.067	0.006
Irritability	0.179	0.071	0.012
Adenoid hypertrophy	0.103	0.052	0.048
Sleep quality	-0.098	0.051	0.053

Table 2. LASSO regression coefficients and statistical characteristics of core features. Note: Negative coefficients indicate that higher feature values are associated with a lower probability of TD comorbid with ADHD; positive coefficients indicate a higher probability. *P*-values were calculated based on bootstrap resampling (1000 iterations) to assess coefficient significance.

Model		Accuracy	Precision	Recall	F1-Score	AUC	PPV	Brier score
Train	Logistic Regression	0.813	0.804	0.813	0.786	0.788	0.820	0.18
	Random Forest	0.831	0.835	0.831	0.804	0.810	0.828	0.15
	Light GBM	0.831	0.822	0.831	0.815	0.819	0.844	0.16
	XG Boost	0.829	0.824	0.829	0.808	0.814	0.835	0.17
	MLP	0.836	0.831	0.836	0.817	0.825	0.841	0.18
Test	Logistic Regression	0.788	0.775	0.788	0.755	0.764	0.798	0.21
	Random Forest	0.793	0.790	0.793	0.753	0.780	0.795	0.18
	Light GBM	0.791	0.774	0.791	0.769	0.777	0.813	0.19
	XG Boost	0.788	0.772	0.788	0.759	0.772	0.803	0.20
	MLP	0.791	0.775	0.791	0.762	0.768	0.805	0.23

Table 3. Performance evaluation of 5 machine learning algorithms.

supplemented, and the results showed that the area under the PRC curve (AUPRC) was 0.76, which verified the effectiveness of Lasso feature screening. 1000 times bootstrap resampling stability analysis showed that the selection frequency of the 7 core features was higher than 85% (learning initiative 98%, guardian type 95%, academic performance 92%, allergic rhinitis 89%, irritability 88%, adenoid hypertrophy 86%, sleep quality 85%), which proved that the feature screening results had good stability (Supplementary Table S3). These seven features were subsequently used for the development of all machine learning models.

Machine learning model performance evaluation

This study follows a structured analytical pipeline: feature screening via Lasso regression, followed by predictive model construction, and finally model interpretation using SHAP. Notably, SHAP was employed solely to explain the prediction logic of the optimal model and was not involved in the feature selection process. Five mainstream machine learning algorithms were employed to systematically develop predictive models for TD comorbid with ADHD and identify the optimal approach: Logistic Regression, Random Forest, Light Gradient Boosting Machine (Light GBM), Extreme Gradient Boosting (XGBoost), and Multilayer Perceptron (MLP).

Among the various performance evaluations, the Random Forest model demonstrated the most outstanding performance: it achieved an area under the curve (AUC) of 0.780 in ROC analysis on the test set, with an accuracy of 0.793, recall of 0.793, and an F1-Score of 0.753 (Table 3). Decision Curve Analysis (DCA) indicated that the model provided high clinical net benefit within the threshold probability range of 10%–30%, while the calibration curve showed strong agreement between predicted probabilities and actual risks. The Brier score of the Random Forest model on the test set was 0.18, indicating good calibration of the model; the Brier scores of other models are shown in Table 3, which are consistent with the calibration curve results (Fig. 3G/H) and all reflect satisfactory calibration performance. The difference in AUC between the training set and the test set is 0.03, which is a slight overfitting and within the acceptable range. This difference is mainly due to the more sufficient fitting of the data distribution by the training set, while the independent test set verifies the generalization ability of the model. The DCA results of different models show that the random forest model has significantly higher clinical net benefit than the logistic regression model in the threshold probability range of 10%–30%, which verifies its advantages in clinical scenarios.

Although all models exhibited a decline in performance metrics on the test set compared to the training set—reflecting expected mild overfitting—the Random Forest model demonstrated the smallest performance degradation. All models achieved an AUC greater than 0.76 on the test set, with Random Forest exhibiting the most robust generalization capability (Fig. 3). Based on its comprehensive advantages in generalization performance and clinical applicability, Random Forest was identified as the optimal predictive model in this study. Although the AUC of the random forest model is slightly different from that of the logistic regression, it

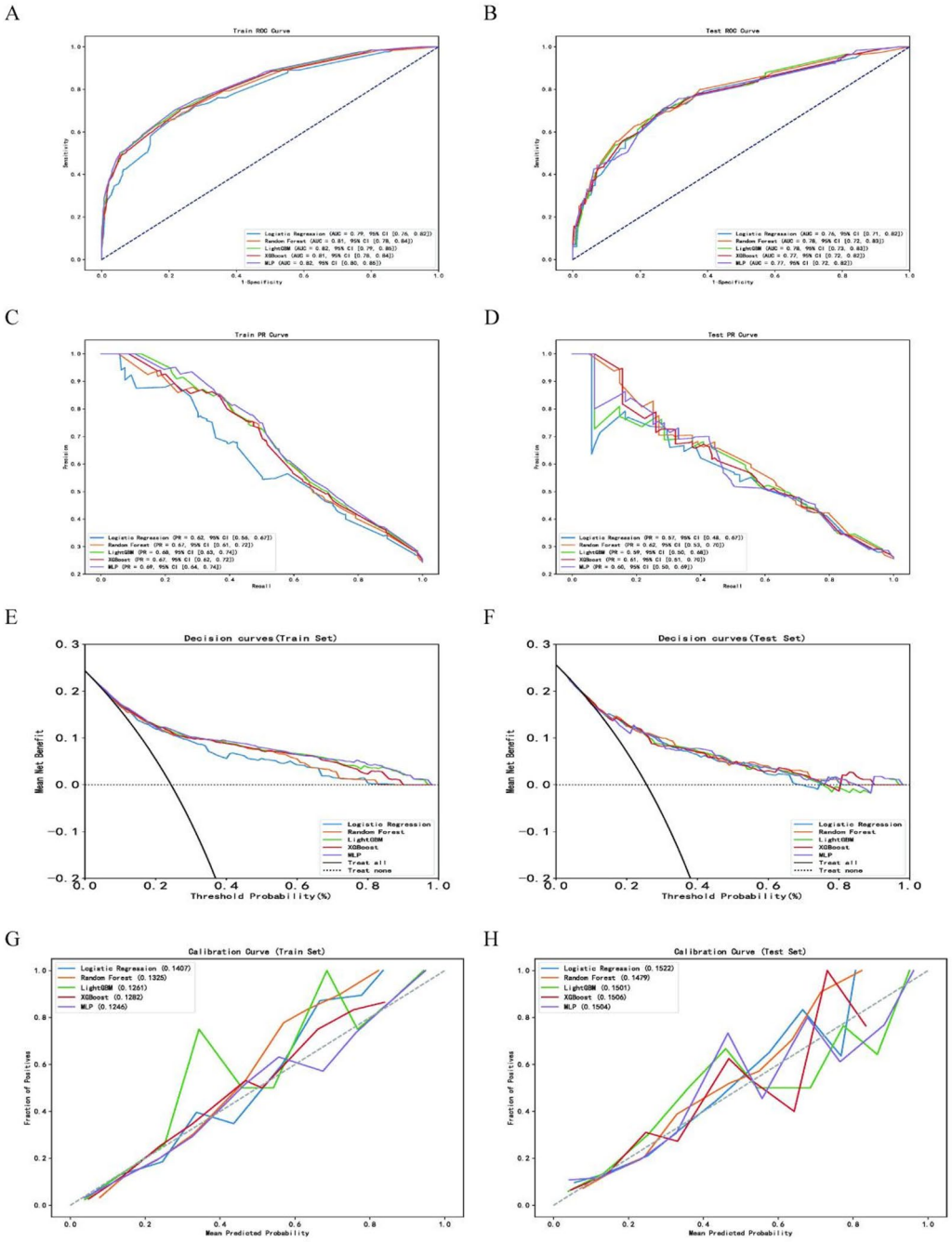


Fig. 3. (A) ROC Curve for the Training Set (B) ROC Curve for the Testing Set (C) PR Curve for the Training Set (D) PR Curve for the Testing Set (E) DCA Curve for the Training Set (F) DCA Curve for the Testing Set (G) Calibration Curve for the Training Set (H) Calibration Curve for the Testing Set.

can capture the non-linear correlation between clinical features (such as the interaction between sleep quality and academic performance), and DCA verification shows that it has higher clinical net benefit, which is more suitable as a clinical screening tool.

SHAP analysis reveals key predictive features

To gain an in-depth understanding of the model's predictive decision logic and enhance interpretability, we employed SHAP to interpret the best-performing Random Forest model. This approach clearly elucidates the influence of variable features on the model's prediction of TD comorbid with ADHD. As shown in Fig. 4A, the summary bee swarm plot comprehensively illustrates the impact of each feature on the model output. The horizontal axis represents the SHAP value—positive SHAP values indicate that a feature pushes the model's predicted probability of TD comorbid with ADHD upward, while negative values push the predicted probability downward. The vertical axis lists the features ranked by their mean absolute SHAP value. The color gradient

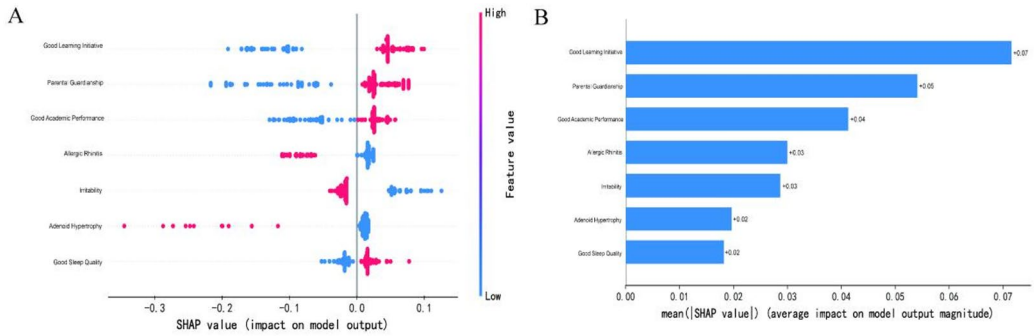


Fig. 4. (A) SHAP beeswarm plot showing the impact direction and magnitude of features on the model's prediction of TD with ADHD (B) Bar plot of mean absolute SHAP values for feature importance ranking.

intuitively reflects numerical differences: red represents high feature values, and blue represents low feature values. Comparison of Lasso regression coefficients and SHAP value directions shows that they are completely consistent: the Lasso coefficients of learning initiative, guardian type, academic performance and sleep quality are negative, and the negative contribution of corresponding SHAP values is dominant; the Lasso coefficients of allergic rhinitis, irritability and adenoid hypertrophy are positive, and the positive contribution of corresponding SHAP values is dominant, which verifies the consistency of model prediction logic.

The analysis revealed significant patterns: High values (red points) of learning initiative, good academic performance, parental guardianship, and good sleep quality were predominantly concentrated on the left (negative SHAP value region), indicating that these features are associated with a lower model-predicted probability of comorbid ADHD. In contrast, high values (red points) of irritability, allergic rhinitis, and adenoid hypertrophy were concentrated on the right (positive SHAP value region), suggesting that these features these features are associated with a higher model-predicted probability of comorbidity.

The summary bar chart (Fig. 4B) displays the mean absolute SHAP values for each feature, with longer bars indicating greater feature importance. Among them, learning initiative (mean(|SHAP|) = 0.07) and parental guardianship (0.05) contributed most significantly to the predictive model, followed by good academic performance (0.04), allergic rhinitis (0.03), irritability (0.03), adenoid hypertrophy (0.02), and good sleep quality (0.02). These mean absolute SHAP values are model-specific relative importance indicators and do not represent absolute risk effects in clinical practice.

To assess the consistency of feature importance across different interpretive methods, we compared the rankings derived from SHAP analysis (based on mean |SHAP value|) with those from the LASSO regression (based on |β|). The results demonstrated strong concordance between the two approaches (Spearman's $\rho = 0.92$, $P < 0.001$). Both methods identified learning initiative and guardian type as the two most influential predictors. Features including academic performance, allergic rhinitis, and irritability were consistently ranked among the top five in both analyses, while sleep quality and adenoid hypertrophy were ranked lowest in both.

In summary, SHAP analysis further emphasized the predictive importance of clinical features such as learning initiative, guardian type, allergic rhinitis, and irritability in the model's prediction of TD comorbid with ADHD. The core role of SHAP analysis is to explain the prediction logic of machine learning models, help clinicians understand the decision-making basis of the models, rather than reveal the clinical or biological mechanisms of diseases.

Sensitivity analysis

To assess the robustness of our findings, we conducted several sensitivity analyses (Supplementary Table S3). First, a feature coding sensitivity analysis was performed by reversing the binary coding of all seven core features (e.g., setting 1 = passive, 0 = active for learning initiative). The Random Forest model retrained on the reversed coding achieved a test set AUC of 0.775 (95% CI: 0.738–0.812), representing a ΔAUC of 0.005 compared to the original coding (0.780, 95% CI: 0.742–0.818). This negligible change indicates that the model's predictions are not materially affected by the choice of binary encoding. Second, to evaluate the impact of class imbalance, we separately applied SMOTE oversampling and random undersampling to the training set before model retraining. The resulting test set AUCs were 0.772 (95% CI: 0.735–0.809) for SMOTE and 0.774 (95% CI: 0.737–0.811) for undersampling, corresponding to ΔAUCs of 0.008 and 0.006, respectively. All AUC differences were below 0.02, confirming the model's stability under different class-balance handling strategies. Third, we estimated the uncertainty of SHAP-based feature importance rankings using bootstrap resampling (1000 iterations). For each iteration, the Random Forest model was refitted on a bootstrap sample of the training set, and mean |SHAP| values were recomputed on the out-of-bag observations. The 95% confidence intervals of the mean |SHAP| values for the seven core features are presented in Supplementary Table S3. Although minor overlaps of the confidence intervals were observed between adjacent features (e.g., guardian type and academic performance), the ranking order of the seven features remained completely stable across all 1000 bootstrap iterations, with learning initiative and guardian type consistently identified as the top two predictors. This high stability supports the reliability of the interpreted feature contributions.

Discussion

The core conclusion of this study is based on the clinical correlation analysis of machine learning models, aiming to identify high-risk groups of TD complicated with ADHD, rather than revealing the causal mechanism of the disease. TD are neurodevelopmental conditions characterized by sudden, repetitive, non-rhythmic motor movements or vocalizations. These symptoms typically emerge during childhood, most commonly between the ages of 5 and 10, with a higher incidence in boys than in girls[23]. In addition to tic symptoms, individuals with TD may exhibit sensory features, specifically premonitory urges and heightened, persistent sensitivity to non-harmful external or internal stimuli[24]. These distinctive perceptual and emotional regulation responses often significantly impact patients' daily lives. Furthermore, TD patients may present with various behavioral comorbidities, including ADHD, obsessive-compulsive disorder (OCD), and autism spectrum disorder (ASD)[25]. Current clinical scale assessments indicate that children with TD frequently present with comorbid ADHD. Although TD and ADHD often co-occur, the nature of the relationship between their symptoms remains unclear[14]. Early identification of ADHD comorbidity in children with TD continues to pose significant challenges in clinical practice. Given these diagnostic difficulties, establishing reliable clinical prediction models is crucial for improving the accuracy and timeliness of comorbidity recognition.

To address the diagnostic complexity of TD-ADHD comorbidity, this study leveraged the capability of machine learning to integrate heterogeneous clinical data by developing a predictive model for ADHD comorbidity in children with TD. Specifically, we employed Lasso regression combined with 10-fold cross-validation for feature dimensionality reduction and selection. Seven core features were ultimately identified as key predictive features for the model's classification of TD comorbid with ADHD: learning initiative, guardian type, academic performance, allergic rhinitis, irritability, adenoid hypertrophy, and sleep quality. Through systematic comparison of five machine learning algorithms (Random Forest, Light GBM, XGBoost, MLP, and Logistic Regression) across performance metrics including ROC curves, precision-recall curves, decision curves, and calibration curves, the Random Forest model was determined to be the optimal predictive model. SHAP summary plots and bar charts highlighted the substantial contributions of features such as learning initiative and guardian type to the model's predictions.

Notably, features such as learning initiative and academic performance may reflect shared underlying constructs related to executive function and motivation. The simultaneous inclusion of both in the model should be interpreted with caution, as their individual importance might be influenced by this conceptual overlap. Nonetheless, from a predictive standpoint, both contributed to model performance, and their combined effect is clinically meaningful for screening purposes. The strong concordance between the feature importance rankings derived from LASSO regression and SHAP analysis (Spearman's $\rho = 0.92$, $P < 0.001$) underscores the robustness of our core predictors—such as learning initiative and guardian type—in identifying TD comorbid with ADHD, irrespective of the interpretative method. More importantly, SHAP analysis provided critical added value beyond the linear, global summaries of LASSO. Specifically, SHAP (1) captured nuanced non-linear relationships and threshold effects (e.g., for sleep quality), (2) offered individual-level explanations that clarify risk attribution for specific cases, and (3) delivered intuitive visualizations that enhance model interpretability for clinical practitioners. Thus, while LASSO confirmed feature selection stability, SHAP significantly extended our understanding of the model's decision logic, thereby improving its clinical translatability and utility in personalized assessment. One notable finding is that the YGTSS total score was not selected by LASSO regression as a core predictive feature. This suggests that, within our cohort, the severity of tic symptoms per se does not independently discriminate between children with TD alone and those with TD comorbid with ADHD. Instead, the model captured behavioral, emotional, environmental, and allergic comorbidity factors. This is consistent with clinical observations that ADHD comorbidity in TD is driven more by shared neurobehavioral dysregulation than by tic severity. Thus, the model should be understood as capturing phenotypic correlates rather than tic-specific clinical features.

The model identified family guardianship patterns and learning behavior characteristics as the most influential predictive features for distinguishing children with TD and ADHD from those with TD alone. Our model identified parental guardianship as a key predictor, indicating that parental guardianship is a key predictive feature in the model; children with parents as primary guardians had a lower model-predicted probability of being classified as having ADHD comorbidity. This should be interpreted as a model-based association rather than a causal protective effect. This association may be explained by more stable emotional support and a structured environment, which are theorized to be conducive to better behavioral regulation. Conversely, we observed that insufficient parental involvement or frequent family conflicts are associated with increased ADHD manifestations, which may be related to neurochemical imbalances and psychological stress[26–28]. Additionally, TD children with high learning initiative and strong academic performance were associated with a lower model-predicted probability of being classified as comorbid with ADHD[29,30]. Baseline data further indicated that in the TD with ADHD group, 45.3% of children exhibited poor learning initiative and 41.7% had below-average academic performance—significantly higher proportions than those in the TD-only group (23.1% and 22.5%, respectively). This underscores the close relationship between academic performance and comorbidity risk. Furthermore, high-quality family interactions, such as parent-child reading activities, have been shown to positively modulate the relationship between tic severity and quality of life[31]. Although the socially constructed predictors identified in this study (such as guardianship type and learning initiative) are subjective indicators, they can be quickly assessed through standardized questionnaires. In the future, these indicators can be integrated into the routine screening scale for children's neurodevelopmental disorders to form a comprehensive screening tool combining 'objective signs + subjective behaviors', thereby improving the efficiency of comorbidity identification[32]. The core predictive indicators identified in this study can provide reference directions for clinical screening, and the intervention value of these indicators needs to be verified by prospective studies in the future.

Moreover, our findings suggest that allergic conditions such as adenoid hypertrophy and allergic rhinitis are important predictive features supporting the classification of ADHD comorbidity in children with TD. While our study did not measure biological markers to elucidate the underlying mechanisms, the observed associations are consistent with previous hypotheses implicating immune dysfunction and neuroinflammation in neurodevelopmental disorders. For instance, literature suggests that adenoid hypertrophy is linked to chronic hypoxia and sleep-disordered breathing, which could theoretically compromise prefrontal oxygen supply and dopamine metabolism, potentially contributing to impaired executive functions[33–35]. Similarly, allergen-activated mast cells are thought to initiate neurogenic inflammation and release pro-inflammatory cytokines, processes which have been theorized to dysregulate key neurotransmitter systems including dopamine and norepinephrine, thereby potentially exacerbating attention and behavioral regulation deficits[36–40]. The predictive correlation between indicators such as allergic rhinitis and adenoid hypertrophy and comorbidity may be related to neurodevelopmental abnormalities caused by chronic inflammation. This hypothesis needs to be verified by prospective studies combining biological markers.

Our results demonstrate that sleep quality and emotional dysregulation serve as significant clinical predictive features for ADHD comorbidity in TD patients. Studies have shown that the prevalence of insomnia in TD patients (32.16%) is significantly higher than in the general population (13.70%), with ADHD comorbidity being one of the key drivers of sleep disturbances[41]. From a pathophysiological perspective, sleep deprivation and fragmentation are hypothesized to exacerbate impulsive behaviors and emotional dysregulation by impairing prefrontal-limbic circuit functionality and reducing serotonergic neurotransmission[42,43]. Meanwhile, irritability, as a core predictive symptom in our model, is known to interact bidirectionally with tic symptoms in TD. This interaction is thought to involve abnormal amygdala connectivity and deficits in cognitive flexibility[44,45], with approximately 20%-40% of TD patients potentially experiencing severe anger outbursts[46].

Notwithstanding the satisfactory predictive performance of our model, this study has several limitations. First, its retrospective, single-center design based on electronic medical records is susceptible to potential biases, including selection bias and information bias due to inconsistent data recording. This study adopted a single-center retrospective design. Some subjective indicators relied on guardians' recall, which may lead to recall bias. Future prospective studies can adopt real-time recording methods to reduce such bias. Second, machine learning models are inherently stochastic; although we used cross-validation, the reported performance and feature importance may exhibit variability. Third, the identified feature importance signifies association, not causation. The predictive features should not be interpreted as direct causal risk factors, as their relationships could be confounded by unmeasured variables. Moreover, several key predictive features—including learning initiative, academic performance, irritability, and sleep quality—conceptually overlap with core or associated symptoms of ADHD. Although we attempted to distinguish baseline academic functioning prior to TD exacerbation, some overlap is inevitable in a retrospective design. Therefore, our model should be interpreted as a clinically informed screening and classification tool that identifies children with TD who are likely to have concurrent ADHD based on currently manifested behavioral patterns, rather than a purely prognostic model for future risk. This distinction is important to avoid circular reasoning. Future studies can include more biological indicators unrelated to ADHD to further optimize the independence of the model. Fourth, the observed slight performance drop in the test set (AUC = 0.780) compared to the training set (AUC = 0.810) may indicate a degree of overfitting, despite our use of feature selection and cross-validation to mitigate this. Fifth, the current model is not intended to replace standard diagnostic procedures. It should be used only as an exploratory screening support tool to help clinicians identify children with TD who may benefit from further ADHD evaluation. Definitive diagnosis must follow DSM-5 criteria and comprehensive clinical assessment. Sixth, and crucially, this model was developed and validated within a single-center retrospective cohort without external validation. The generalizability and transportability of our findings, particularly for the family and social variables, across different healthcare settings, regions, or populations remain uncertain. Therefore, the predictive model developed in this study should serve as an ancillary tool to be integrated with comprehensive clinical assessments. It cannot replace the DSM-5 diagnostic criteria and clinical scale evaluation. The final diagnosis needs to be determined by specialists based on the comprehensive clinical manifestations of the children. It should be emphasized that these findings are derived from a single-center retrospective study without external validation. The proposed model currently serves as a research tool with potential clinical relevance, not a ready-to-use decision aid. Prospective studies in diverse populations are necessary to confirm its utility.

Conclusion

In summary, we have successfully developed a machine learning model based on clinical practice data to distinguish children with TD comorbid with ADHD from those with TD alone. Among the five machine learning algorithms evaluated in this study, the Random Forest model demonstrated the most outstanding performance. By applying the SHAP method for model visualization, we not only clearly illustrated how various features contribute to prediction outcomes but also effectively determined feature importance. SHAP analysis identified seven key features contributing to the model's prediction probability of TD comorbid with ADHD: learning initiative, guardian type, academic performance, allergic rhinitis, irritability, adenoid hypertrophy, and sleep quality. This comprehensive research approach provides valuable insights into the clinical correlates associated with this comorbid condition, although the model requires external validation before being considered for routine clinical use.

Data availability

De-identified participant data and the study protocol are available upon reasonable request to the corresponding author, subject to approval by Guang'anmen Hospital Ethics Committee. A data dictionary will be provided. Access criteria include approval of a research proposal and signing a data access agreement.

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Author contributions

Z.C. conceptualized the study, drafted the manuscript, and supervised the project; X.W. and Y.C. collected data and performed statistical analyses; Y.H. and R.J. developed and validated the machine learning models; F.H. critically revised the manuscript and provided funding support. All authors approved the final version.

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Competing interests

The authors declare no competing interests.

Ethics approval

This retrospective observational study was conducted in strict accordance with the ethical principles outlined in the Declaration of Helsinki. The study protocol was reviewed and approved by the Medical Ethics Committee of Guang'anmen Hospital, China Academy of Chinese Medical Sciences (Approval No. 2023-211-KY-01). Written informed consent was obtained from all legal guardians of the participants after they had been fully informed of the study's purpose, procedures, and their rights. All personal information collected was rigorously anonymized and stored in a secure environment to ensure the privacy and data security of the participants.

Additional information

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Correspondence and requests for materials should be addressed to F.H.

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